Influencing User Decisions: Dilemmas in Designing Online Interfaces

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ABSTRACT

The choices designers make about user interfaces can influence the choices end-users make when using those interfaces. These steering qualities of design have received much attention in the context of nudging and manipulative design (e.g., so-called 'dark patterns'). This paper discusses ethical design practices and reviews the differences between nudging and manipulative design strategies, focusing on aspects that make them (un)acceptable. We show that existing distinctions are inconsistent and sometimes contradictory, often offering useful but partial or hard-to-follow advice. We illustrate these points with three concrete dilemmas that well-meaning designers face. Our review reveals that nudging and manipulative design both treat users as irrational and plagued by cognitive biases, which can be mitigated or exploited. We question this view and propose developing new interface design principles that treat users as rational and help them make their own deliberate autonomous decisions.

CCS CONCEPTS

• **Human-centered computing → Interaction design theory, concepts and paradigms**; *HCI theory, concepts and models.*

KEYWORDS

Nudge, Dark Patterns, Deceptive Design, Manipulation, Persuasion, Bias, Autonomy, Fairness, Ethical Design

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1 INTRODUCTION

In today's society, everyday activities, such as shopping, dating, and reading the news, are regularly carried out online via software systems that present various options to their users via user interfaces. A growing concern is that user interfaces are not neutral, i.e., they can influence users' behaviors [2, 49, 52, 53] and, e.g., cause users to carry out actions they regret or that do not align with their true objectives [25].

As "there is no neutral way to present choices" [2, 53], and "every design choice is a nudge" [2, p. 44:27] a question designers face is how to nudge in a responsible manner, i.e., how to steer users ethically. While some guides for ethical design in general [e.g., 1, 36] and ethical nudging in particular [e.g., 2, 29] exist, designing fair interfaces is not trivial, leading designers to create potentially manipulative designs by accident [10]. Avoiding such problems requires an understanding of what distinguishes acceptable steering from unacceptable steering. We set out to develop such an understanding in this late-breaking work. However, instead of definite answers, we discover a variety of dilemmas designers can face when trying to design interfaces ethically and responsibly.

In recent years, the steering qualities of interfaces have received increasing attention from the HCI community and related disciplines, with much focus on two — seemingly — distinct general types of steering. The first type comes in the form of so-called nudges. The concept of *nudging* was popularized by Thaler and Sunstein [49], and it traditionally refers to influencing "people's behaviour in a predictable way without forbidding any options or significantly changing their economic incentives" [49]. Decisions encouraged through a nudge should improve people's lives and make them better off, as judged by themselves [49]. Nudges can exist both physically and in the digital domain. Examples include designs that guide users to make a more privacy-aware choices [35] or smart energy-consumption-meters that steer users to use less energy [53].

The second type of steering is known under terms such as deceptive design and dark patterns. These lead to users making decisions that are not in line with their preferences [25] and manipulate users into taking certain actions that typically benefit the party/entity behind the design. Such designs became popular as 'dark patterns' in 2010 via a website that collects said tricks [8] and since have become an active HCI research field [e.g., 20]. Recently, the terminology has changed to be more inclusive and avoid racial connotations [14], resulting in terms like deceptive patterns [7] and manipulative patterns [24]. Examples of manipulative designs include cookie popups that make it easier to agree than to disagree [21], deceptive (fake) countdown timers that pressure users to place an order on a webshop [50], or making the cancellation process of newspaper subscriptions unnecessarily difficult to keep users subscribed [42].

Deceptive design and nudges have a lot in common, as both rely on human biases [18] and employ similar strategies, such as providing default settings [9] or framing [2] to steer users. In fact, similar examples, such as increasing users' donations, have been provided both as a case of manipulation [11] and as an example for a nudge [34]. Yet, in existing literature, nudging is often considered positive and acceptable [34, 47], whereas deceptive design patterns are known as unethical [9, 34]. Furthermore, both are often

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considered to be easily distinguishable. For instance, Özdemir [34] claim there is a "distinctive property which separates nudges from dark patterns", citing Sunstein [44] that "nudges are generally designed to make people better off, as judged by themselves" adding that "dark patterns trick users" [34]. In light of this, the difference between nudges and deceptive designs might tell us more about what distinguishes acceptable from unacceptable steering and give valuable guidance to designers about how to design — if not neutral — then responsibly.

2 DIFFERENCES BETWEEN ACCEPTABLE AND UNACCEPTABLE STEERING

To clarify what exactly makes steering (un)acceptable, this section reviews definitions of what generally is considered acceptable steering (e.g., nudging) and unacceptable steering (e.g., manipulation, deceptive design) and literature on their differences (overview available in Table 1 in Appendix A).

While our review of relevant literature is not exhaustive, we can identify two underlying recurring concepts, namely autonomy and bias. Autonomy is mentioned as a core value that is threatened by online manipulation [46]. The concept is used to question whether nudges are acceptable ("Do nudges compromise people's 'autonomy'?" [40, p. 337]) and as a normative lens to evaluate deceptive designs [3]: "an assessment of whether a design pattern qualifies as 'dark' should account for the sense in which autonomy is threatened" (p.1). The second core concept, cognitive biases, is presented as "systematic errors in judgments and behaviors" [2, p. 44:6], i.e., a human condition that limits people's decision-making capabilities. Bias may steer users [e.g., 9, 18, 38], and as users are affected by their biases, these can either be exploited (manipulation) or corrected/mitigated or used for guiding users (nudging) [2].

Furthermore, we identify three factors that are used to distinguish (un)acceptable steering.[1,2] A first factor is the underlying *intention* of the designer or choice architect. For instance, van Nimwegen et al. [51] claim: "To distinguish the two, we can look at the intentions of the designers. In persuasive and nudging practices designers aim to encourage users to freely explore content and take actions. [...] In deceptive practices on the other hand, designers aim to either trick users into taking actions, or prevent them from performing them" (p. 2). Similarly, in the context of privacy, it is stated that manipulation occurs when "the good intents of privacy nudging are replaced with malicious intents" [9, p. 247]. Conversely, Susser et al. [45] argue that manipulation can also occur if well-intended, as it still alienates users from their decision-making power.

The second factor concerns the qualities of the actual design, i.e., the processes used to steer people. Specifically, nudges are said to be transparent and "maintain or increase freedom of choice" [49, p. 5]. Manipulative design, conversely, is said to trick, mislead, coerce, steer, or deceive users, as collected and consolidated by Mathur et al. [27].

[1] Clausen et al. [12] make a similar distinction and identify (1) intentions, (2) strategies, (3) outcomes, and (4) process as crucial in the ethical debate around designing persuasive systems.

[2] The following distinction between intentions and outcomes is not always clear, as it is sometimes open if *intended* or *actual* outcomes are meant.

The third factor pertains to the outcomes or effects of a nudge. Here, some researchers focus on *whether* the design has an influence, while others focus on specific types of effects, such as an increase in users' welfare. For example, Wilkinson [54] focuses on effectiveness as such and states, "If and when nudging[3] does not succeed in altering behaviour, it is not manipulation" [54, p. 347]. Consequently, if a user is aware of a design's influence and therefore manages not to be tricked, this design would not count as manipulative. A similar view is embedded in the Unfair Commercial Practices Directive, where practices are judged based on whether they are likely to change a consumer's preference [16]. For those patterns that actually have an effect, distinguishing qualities can be 'who benefits' (or is intended to benefit) from the design and the nature of the effect. Generally, choices supported by nudges are expected to be in line with users' preferences or benefit users' welfare [e.g., 3, 13, 49]. In addition, some argue that nudges can also support choices in the interest of societal welfare [e.g., 34]. Deceptive design, in contrast, is said to go against what users want and harm them [e.g., 10, 30, 35], and are reported to benefit the company behind the design, e.g., increase profit [e.g., 27].

3 DESIGN DILEMMAS

As we have seen, various — sometimes complementary, sometimes diverging — views on what distinguishes acceptable and unacceptable steering exist. In the following, we zoom in on (the interplay between) the identified factors in more detail and explore dilemmas that emerge when aiming to apply insights from existing literature in practice.

3.1 The 'manipulation for good' dilemma

As we have seen above, designs are sometimes judged based on the designer's intentions [e.g., 9, 51], on the mechanisms employed [e.g., 27], or on the outcomes [25]. One problem designers face is that one and the same design can be considered acceptable, depending on which (combination) of these factors/criteria are employed. An example could be a health nudge that promotes a healthy salad over a greasy burger, with the users' best interests in mind, but with the shaming slogan 'No, I don't care about my health' accompanying the choice for the burger. This design could be considered acceptable when merely judged based on the designer's intentions. Also, when applying common criteria like transparency, freedom of choice, and whether the nudge benefits the users' welfare [49], such a design would be considered a nudge rather than a deceptive design. Yet, it also falls into the deceptive design category 'confirmshaming', which uses derogatory or condescending words to make the user feel guilt or shame and ultimately persuade them to make a particular decision [8]. Depending on the user, the rather aggressive slogan might hurt them emotionally or amuse them. Similarly, the design might harm a user [27] on one level while contributing to their well-being on another level (e.g., the burger may not be healthy but satisfy them emotionally), raising the additional challenge of how to measure benefit and evaluate desired outcomes, as discussed by Acquisti et al. [2].

[3] In this reference, the term nudging refers to steering rather than to nudging in the sense of Thaler and Sunstein [49].

This example shows that — depending on which criteria are used — a design can qualify both as a nudge and manipulation (or, a nudge can employ a deceptive pattern). Likewise, in terms of *perceived* acceptability, perceptions might differ based on who judges the design. For the designer, the dilemma mostly comes down to the old question of whether the (intended) ends justify the means. As Susser et al. [46] put it: "One can imagine cases where the harm [manipulation causes] to autonomy is outweighed by the benefit to welfare. (For example, a case where someone's life is in immediate danger, and the only way to save them is by manipulating them.)" (p. 11). Therefore, designers face the challenge of steering people in a manner that is proportional to the (intended) benefits [2] and the dilemma of balancing harm and benefits for users and society.

3.2 The 'who benefits' dilemma

As discussed in Section 2, one factor that is considered to play into the acceptability of a design is its effect/outcome. When trying to design responsibly, one aims for 'good' outcomes. However, there is a potential tension between steering towards decisions that contribute to the users' welfare [9] versus those that benefit society [40]. In other words, whether a design is considered acceptable can depend on which factor (user welfare or societal welfare) is considered.

Of course, as users generally are part of society, a certain choice might contribute to both the users' and society's welfare. An example is saving money on energy and being more sustainable [40]. However, at other times, the two goals might not align, like in the case of donations, which would increase societal welfare but could financially harm users [11, 34]. Thus, the designer faces the dilemma of acting either in the users' or society's interest and/or balancing both. (This, of course, is even more problematic as users are not a homogeneous group and have diverse perceptions of 'good' [39], and as the benefits for users and society might be difficult to predict.)

When societal and end-user welfare are not aligned, it is unclear how to weigh end-users' interests against those of society at large. While traditional design methods, like human-centered design, emphasize user needs, more recent developments like humanity-centered design [33] show a shift towards a more societal perspective.

3.3 The 'too much of a good thing' dilemma

The processes and mechanisms used to steer people are also considered to distinguish acceptable and unacceptable forms of design (see Section 2). In this context, transparency and freedom of choice are two of the main characteristics of a design that is said to avoid manipulation [49]. By being transparent, the user is at least aware of the intentions of a design and, therefore, can decide more consciously whether or not to make the intended decision. Similarly, preserving freedom of choice means the user is also free in the decision to go against the nudge. However, there are some problems with these concepts.

When it comes to transparency, it is unclear how and whether *actual transparency* about choices and their consequences can be achieved in practice, considering the complexity of many decisions and the time and cognitive effort users can realistically spend on them. Designing for transparency is challenging because *being more transparent (e.g., about details) can decrease actual transparency*. To illustrate this, imagine a cookie popup asking a user to accept third-party cookies, with the simple question "Do you accept third-party cookies?". This arguably will not sufficiently inform a user with little knowledge about cookies. The other extreme — e.g., explaining cookies in detail, listing potential harms and benefits, and giving insight as to why one of the two buttons in the popup is made more visible — would also not be *actually* transparent, as the amount of information would be infeasible to process. This transparency problem can be witnessed in, e.g., the context of privacy disclosures, where reading the notice of each website one visit once is estimated to cost users 40 minutes per day [28]. Considering this, skipping such notices can arguably be considered rational (see Nehf [31], who make a similar argument in the context of terms and conditions). The problem with getting transparency right is also raised by Nissenbaum [32, p. 36], who have introduced the so-called transparency paradox, which Susser et al. [46] summarize as follows: "we are bound to either deprive users of relevant information (in an attempt to be succinct) or overwhelm them with it (in an attempt to be thorough)" (p. 13).

A similar problem occurs when aiming to preserve the users' freedom of choice. The absence and obstruction of certain choices are seen as problematic, e.g., as in the case of the deceptive design "forced action" [35]. Thus, freedom of choice clearly seems desirable. However, a challenge lies in how many and which exact choices to provide. In fact, it has been argued that the "[t]he value of choice in itself may depend on culture, and even in cultural contexts that value choice, too much choice can lead to paralysis, bad decisions, and dissatisfaction with even good decision" [41, p. 106].

Consider an extension of the earlier cookie pop-up, where a list of thousands of partners is provided to indicate to the user with whom their data can be shared accompanied with an individual toggle for each partner that needs to be adjusted. While this arguably presents users with freedom of choice on a superficial level, the fact that so many choices need to be made essentially means that they cannot make a meaningful choice. Given that users' time and resources are limited, it may be rational not to spend too much precious lifetime on such choices. Hence, just like being transparent about things does not necessarily lead to transparency, providing choices does not necessarily lead to freedom of choice. As Acquisti et al. [2] put it in the context of privacy: "too much information and too many settings can be overwhelming" (p. 44:11).

For designers, this means they need to find a balance between *information transparency and overload* as well as between *freedom of choice and freedom from choice* (freely based on the two famous concepts of liberty proposed by Isaiah Berlin). They need to identify which information is relevant to users, which choices are meaningful, and on which level of detail or granularity choices and information should be offered (as also suggested by Terpstra et al. [48]).

4 CLOSING REMARKS AND FUTURE WORK

The choices designers make about user interfaces can, in turn, influence the choices, values, and beliefs of end-users [3, 6] and shape the way the user perceives the world and their reality [52]. We have set out to find answers on how to design responsibly, given that no design is neutral [2, 53]. Looking at practices where steering is explicit, deliberate, and directional, such as nudging and deceptive design, has not led to definite answers but has revealed a series of design dilemmas. In the following, we present and discuss six preliminary conclusions and ideas for future work.

First, there is no clear consensus on what distinguishes nudging from manipulation. Depending on which criteria are applied to categorize a design, one and the same design can be considered a nudge and a deceptive design. The view of nudging as an acceptable practice and the use of deceptive design as an unacceptable practice has been challenged, e.g., by the possibility of implementing classical deceptive designs to benefit the user or society. The fact that the same design strategy (e.g., defaults and framing) can be used both for sinister purposes as well as to protect users [13] means that a design cannot be judged by solely looking at the design elements used (as also suggested by Hoepman [23]). Future work is needed to better disentangle when a practice is a nudge or a deceptive design and when both can be considered unacceptable or acceptable. Follow up research can also broaden the scope of our investigation to 'more neutral' (less directional) types of steering that would not necessarily be considered nudging or manipulation. In addition, existing work on ethical design [e.g., 4, 36], ethical nudging [e.g., 29, 37], and inclusive design [e.g., 1] could be incorporated. For instance, we expect that valuable existing guidelines for specific contexts, such as guidelines for ethical privacy and security nudging [2], could be generalized so that they are applicable more broadly.

Second, whether a design fulfills certain criteria (e.g., being transparent, benefiting the user, offering freedom of choice) often depends on the person interacting with the design and the individual's background, values, and beliefs. A logical response to this could be to consider the personalization of user interfaces [15], e.g., to better align provided information with users' needs and provide (more) meaningful choices. However, personalization is also deeply problematic, as the personal data obtained for this also opens up new possibilities for manipulation, e.g., manipulation that exploits individual decision-making vulnerabilities [46]. This 'personalization dilemma' can be witnessed in current societal debates. Whereas personalization is seen as a worrying trend for deceptive design/manipulation that would increase harm to users [30], sustainability debates view personalization as a potential solution and, e.g., consider making people's first flight cheaper than their fifth flight. In our opinion, the potential benefits and harms of personalization in the context of ethical and responsible user interface design are an exciting area for future research.

Third, opinions on whether a design fulfills certain criteria can vary in a diverse society. The 'traditional' view, proposed by Thaler and Sunstein [49] is that a nudge makes people better off, as judged by themselves. However, even if interfaces are *intended* "to make people better off, as judged by themselves" [44, p. 1], it is often the designer who *actually* judges what will make people better off. In the digital domain, this is often a judgment for users from diverse backgrounds and cultures, possibly leading to design decisions that will benefit some users at the expense of others. In our opinion, the only valid approach to the ambition of "making people better off, as judged by themselves" is to involve as many potential and *diverse* end-users of the design in the design process, accounting for vulnerable groups and deliberately involving people with diverging interests and believes. As a next step, a qualitative study could explore diverse users' perspectives on the acceptability of various types of steering across different contexts (e.g., donations, health, privacy), extending previous user perception research on persuasive and manipulative design [e.g., 5, 19, 26].

Fourth, both nudging and manipulation literature often assume users to be irrational and plagued by cognitive biases that either can be exploited (manipulation) or that need to be mitigated (nudging). However, there is a growing body of literature that argues that the influence of biases is overrated and that human biases are rational as they provide tools to deal with uncertainty [18] and limited time and cognitive capacity [43]. Hence, we believe it would be worthwhile to approach design-based steering through this critical lens and reconsider whether — as argued in other contexts — "fine-tuned intuition is mistaken for bias" [18, p. 315] when users interact with deceptive designs and nudges. Accordingly, we propose to develop new interface design principles that are rooted in treating users as rational and that aim at helping them make their own deliberate autonomous decisions (e.g., 'boosting' [22]). To develop the principles, we suggest adopting a value-sensitive approach that combines the values of diverse end-users with public values, e.g., focusing on autonomy rather than traditional HCI goals like efficiency.

Fifth, achieving design goals, like providing transparency and freedom of choice, is challenging, if not impossible. This, however, does not mean designers should not strive to do so. We think turning the focus towards responsible design processes (e.g., involving diverse users) rather than design goals (e.g., transparency) might be a way forward.

Sixth, this work has focused on acceptable forms of steering. In the future, it would be desirable to explore under which conditions users perceive steering not just as acceptable but rather as beneficial or desirable.

As this is a work in progress aimed at fostering discussion, neither our review of existing literature nor our list of dilemmas is exhaustive. Yet, we believe our work provides a valuable road map and possible entry points to further study how to account for the steering nature of interaction design and user interfaces.

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A THEMES IN LITERATURE

		Concepts		Acceptable steering				Unacceptable steering						
		autonomy	bias	good intentions	users' preferences/welfare	societal welfare	Transparent, Freedom of choice	Bad intentions	harms users/against their interest	company's interest	non-transparent, restrictive	needs to be effective	intentional, regardless of intention	regardless of outcome as judged by user
Phil, Psych	Gigerenzer [17]	●	●		●		●							
Phil	Schmidt and Engelen [39]	●	●								●			
Phil, Tech	Ahuja and Kumar [3]	●	●	●	●		●	●	●	●	●			
Tech	Özdemir [34]		●		●	●	●	●	●		●			
Tech	Mathur et al. [27]	●	●					●	●	●	●			
Tech	Chamorro et al. [10]	●			●		●		●	●	●		●	
Tech	van Nimwegen et al. [51]	●		●	●		●	●	●	●	●			
Tech	Narayanan et al. [30]	●	●		●				●	●				
Tech	Costello et al. [13]		●		●				●	●				
Tech	Acquisti et al. [2]	●		❍	●	●	●		●		●			
Tech	Bösch et al. [9]		●	●	●			●	●	●	●			
Law, Tech	Potel-Saville and Francois [35]	●	●	●	●	●	●	●	●	●	●			
Law, Tech	Susser et al. [45]	●	●										●	●
Law	Luguri and Strahilevitz [25]	●	●		●	●			●	●		❍		
Law	EUR-Lex [16]		●					●	●		●	●		
Law, Econ	Thaler and Sunstein [49]	❍	●		●	❍	●							
Econ	Schubert [40]	●			●		●			❍				
Politics	Wilkinson [54]	●	●									●	●	●
Politics	Schmidt [38]	●	●		●	●	●			●				
	Intention			▲	▲	▲		▲		▲			▲	
	Process						▲				▲			
	Outcome				▲	▲			▲	▲		▲		▲

Table 1: Concepts and factors found in literature from different fields (mentioned: ●, implied: ❍, related to factor: ▲)